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INTELLIMAIL

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AI POWERED INBOX MANAGEMENT WEB APP

PROBLEM

Individuals often juggle multiple communication platforms, including personal and institutional/work email accounts, messaging applications such as WhatsApp and Instagram, resulting in fragmented communication and reduced productivity. Constant switching between apps, cluttered inboxes, and hidden important messages contribute to overload, missed deadlines, and poor time management.

SOLUTION

Intellimail proposes a centralized web app that consolidates multiple email and messaging platforms into a single interface, enhanced with RAG and AI agents that act as a personal assistant that intelligently process, summarize, and respond to messages, in order to enhance productivity.

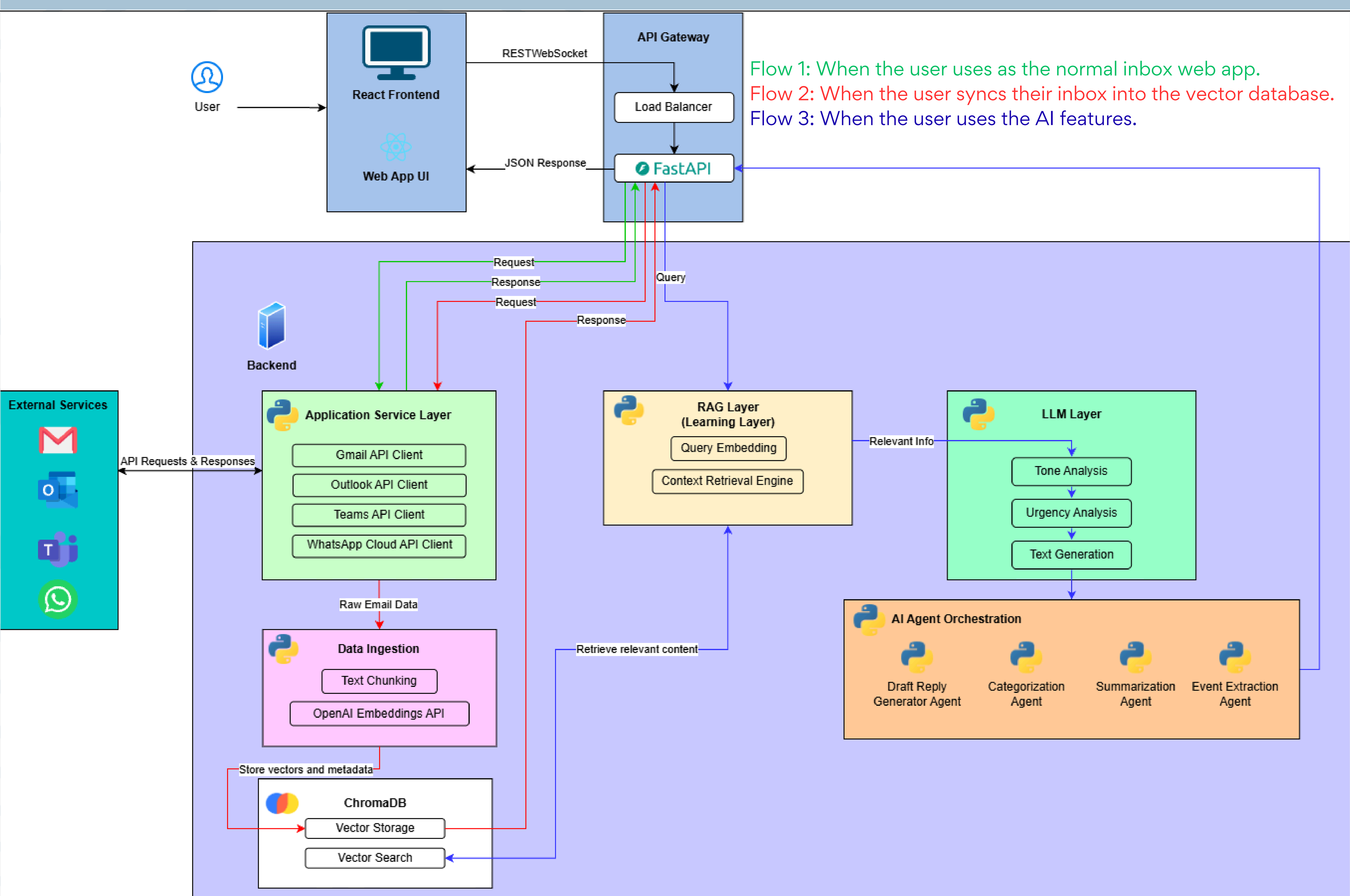
HOW IT WORKS

- **Connect**: User links their accounts through secure OAuth authentication.
- **Unify**: All messages from every connected platform are pulled into a single, organized inbox.
- **Sync**: The app syncs the inbox, converting inboxes into a searchable AI knowledge base.
- **Analyse**: Multiple AI Agents automatically summarize, categorize, prioritize and scan for events across all messages/inboxes.
- **Interact**: Users can chat with the AI living in the web app such as asking questions regarding their inboxes, finding emails, or drafting replies.

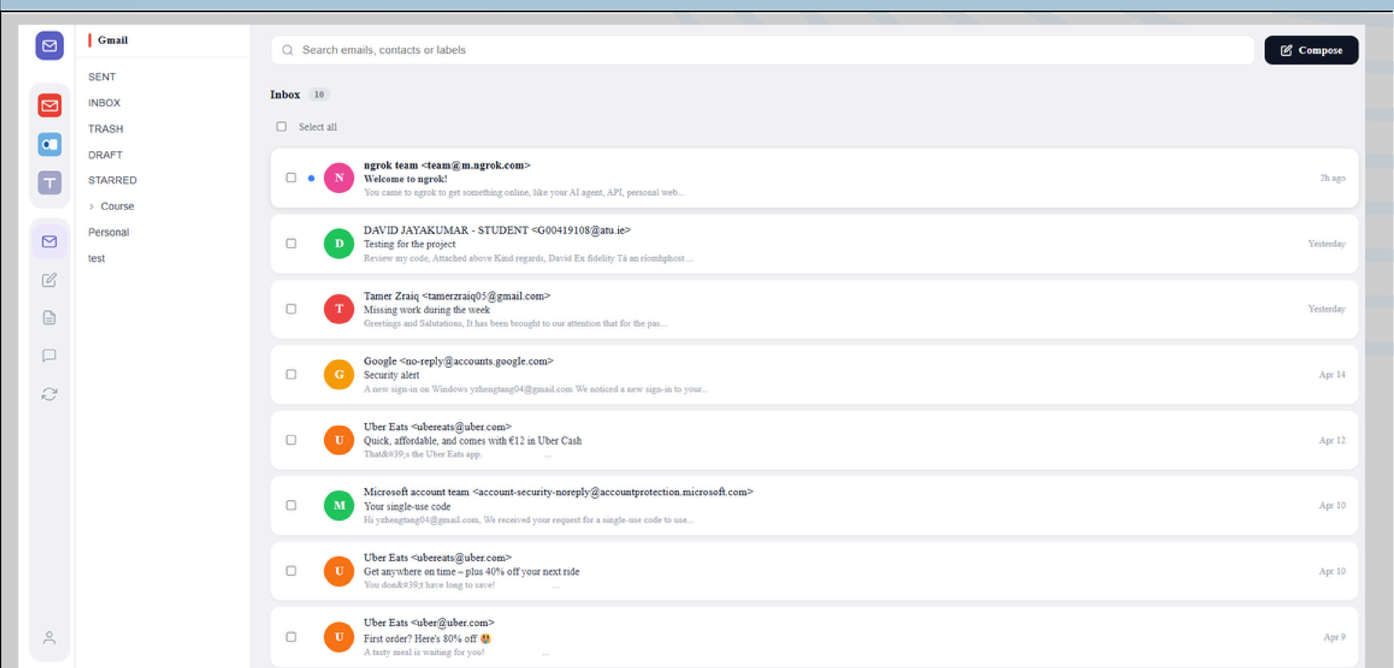
Using React and Typescript for frontend that communicates with Python backend services, user connects their social accounts through OAuth authentication. This allows the web app to fetch and manage inboxes through the API service layer. The inboxes are processed through an AI ingestion pipeline that converts into semantic vectors that is stored in the vector database.

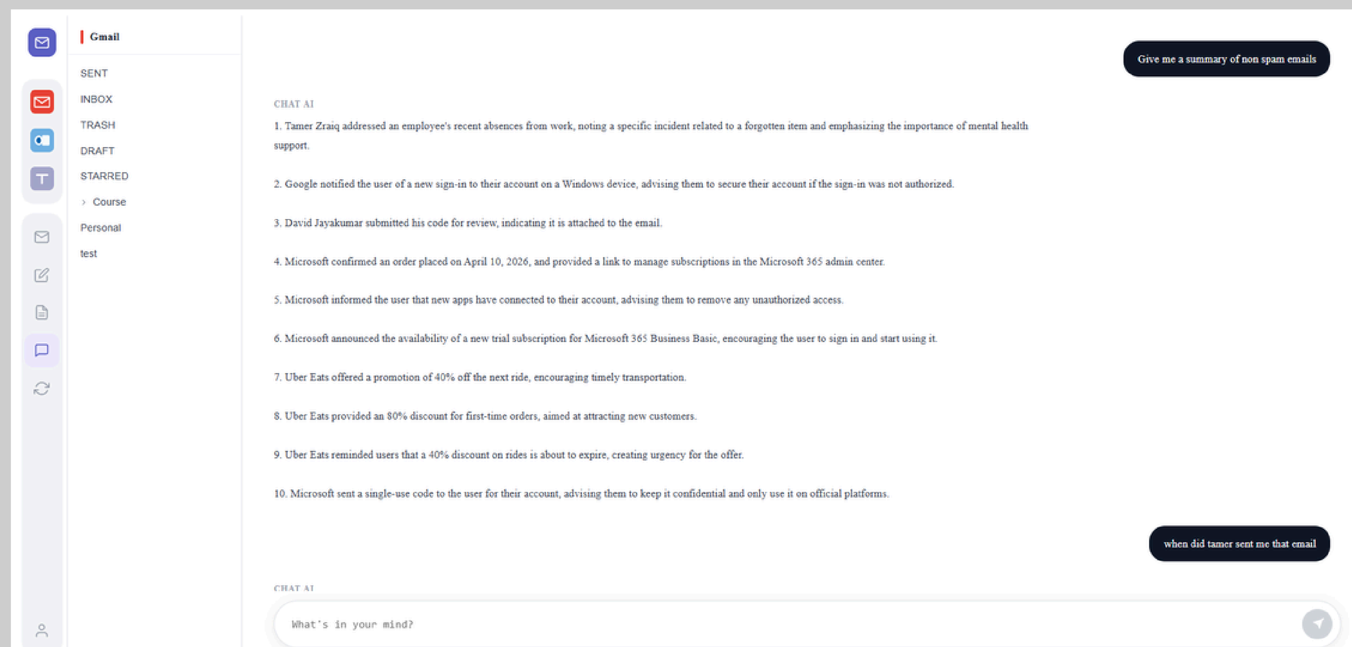
When AI features are triggered, relevant datas are retrieved using RAG and passed as context which powers the AI agents that are responsible for different tasks.

ARCHITECTURE DIAGRAM



RESULTS





A fully working unified inbox web app where user can navigate through different communication platforms, and also an AI chat page.

WHAT IS RAG AND WHY?

Retrieval Augmented Generation (RAG) is a technique that enables LLMs to retrieve and incorporate new information from a specific data sources. With RAG, LLMs will refer to a specified set of documents, then respond to user queries.
This project uses RAG so that the AI agents can generate responses grounded in the user's actual emails rather than relying solely on GPT's general training data, ensuring the answers are accurate, relevant, and specific to the user's inbox.

TECHNOLOGIES USED

React
ChromaDB
MSGraph API
TypeScript
FastAPI
AXIOS
Gmail API

SCAN ME